

Learnify: A Secure and Dynamic Online Quiz Platform

The project aims to provide an efficient, flexible, and secure environment for conducting online quizzes. In today's era of digital learning, such platforms play a crucial role in assessing knowledge and skills effectively. This system is developed using the MERN (MongoDB, Express.js, React.js, and Node.js) stack, which ensures a modern, responsive, and scalable architecture.

The platform has two types of quizzes: default quizzes and customized quizzes. The default quizzes include various aptitude-based tests such as verbal, logical reasoning, arithmetic, and quantitative aptitude. Each quiz is time-bound and automatically submitted once the timer expires, ensuring fair and efficient evaluation. The customized quiz feature allows users to create their own quizzes by defining questions and answers according to their requirements. Once a customized quiz is created, the system generates a unique access code that can be shared with other users or students. This code enables them to attempt the quiz within the predefined start and end time, maintaining control and authenticity of the test.

To ensure data security and integrity, the system implements JWT (JSON Web Token) based authentication, OTP-based email verification, and encryption-decryption of database records. These mechanisms collectively prevent unauthorized access and protect user credentials and quiz data from potential security threats. The platform thus provides a reliable and user-friendly solution suitable for both educators and learners, combining accessibility, security, and performance in a single system.
